DSN AI Bootcamp 2026 Project Participation – Data Analytics Track

Answer & Submission Sheet

Participant Name	**Bashir Lawan Isyaku**
Dashboard tool used	**Power BI**

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Part A - Data Cleaning Review

Identify at least three meaningful data-quality issues you found. For each issue, explain what you found, what you did, and why your action was appropriate.

Data-quality issue	What you found	What you did	Why this action was appropriate
Duplicate Transaction IDs	*The dataset initially contained 12 duplicated Transaction IDs. After comparing the duplicated records, I found that the duplicate rows contained the same information as their corresponding original records.*	*I removed the 12 duplicate records, reducing the dataset from 432 rows to 420 unique transactions.*	*Because the duplicated records contained identical transaction information, keeping both would have counted the same transactions twice and inflated metrics such as revenue, units sold, transaction count, and profitability. Removing the duplicate records ensured that each transaction was represented once.*
Missing Cost_Per_Unit_NGN values	*There were missing values in the Cost_Per_Unit_NGN column. After investigating the dataset, I found that cost per unit was consistently associated with the corresponding product.*	*I mapped the missing cost values to the corresponding product's established cost per unit.*	*Cost per unit is required to calculate profitability. Since the dataset showed that the cost was determined by product, using the established cost associated with the same product provided a consistent way to recover the missing values rather than arbitrarily replacing them with an overall mean.*

Revenue calculation inconsistencies	*The reported revenue did not agree with the independently calculated revenue for 22 transactions. The validated calculation was:* *Revenue=Units×Unit Price×(1−Discount)* *The calculation matched **398 of the 420 transactions**, approximately **94.8%** of the records.*	*I created Revenue_Calculated_NGN, Revenue_Difference_NGN, and Revenue_Ratio columns to validate the reported revenue. After investigating the 22 discrepancies and finding no consistent business pattern explaining them, I used the calculated revenue as Revenue_Final_NGN for subsequent analysis while retaining the original reported revenue for comparison.*	*Using a consistently validated calculation provides a reproducible basis for revenue analysis. Retaining the original reported revenue also preserves the source value for audit and comparison.*

Which data-quality issue had the greatest impact on your analysis?

*However, based on what we've found so far, the **revenue inconsistency is a strong candidate**, because revenue affects almost every major sales-performance metric.*

Explain how correcting that issue changed or affected the analysis.

***The inconsistent reported Revenue field had the greatest potential impact on my analysis.** Revenue is a fundamental measure used to evaluate sales performance, products, markets, channels, and trends. Although most records followed the validated revenue formula, 22 transactions contained discrepancies. Using a consistently recalculated revenue measure ensured that the subsequent analysis and dashboard were based on a reproducible calculation rather than potentially inconsistent reported values.*

Part B - Dashboard Audit

For each management claim, state your verdict, provide the corrected result where applicable, and explain the evidence supporting your conclusion.

1. Abuja is the most profitable market.

Verdict

Incorrect

Corrected answer / result

Lagos is the most profitable market, while Abuja has the lowest profit among the five markets.

Explanation / evidence

*Lagos generated approximately **₦13,692,700 in revenue and ₦4,579,900 in profit**. Abuja generated approximately **₦6,684,200 in revenue and ₦1,698,100 in profit.***

The state-level analysis shows that Lagos ranks highest in revenue, profit, units sold, and transactions. Abuja ranks lowest in revenue, and profit. Therefore, the original claim that Abuja is the most profitable market is contradicted by the cleaned analysis.

2. Electronics is the strongest-performing product category and should receive more budget.

Verdict

Misleading

Corrected answer / result

Electronics is the strongest-performing category by the key financial measures in this dataset, generating ₦29,346,500 in revenue and ₦8,986,500 in profit. However, the data alone does not establish that it should automatically receive more budget.

Explanation / evidence

Electronics generated the highest revenue, profit, units sold, and transactions among the categories *analyzed. Home followed with ₦13,206,300 revenue and ₦3,674,300 profit, while Accessories recorded ₦4,126,925 revenue and ₦1,929,625 profit.*

However, "should receive more budget" is a management decision that would require additional information such as budget requirements, growth potential, return on investment, inventory constraints, and future demand. Therefore, the performance part of the claim is supported, but the budget recommendation is not established by this dataset alone.

3. Higher discounting is improving profitability.

Verdict

Incorrect

Corrected answer / result

Higher discounting is associated with lower profitability in the analyzed data.

Explanation / evidence

*The 0–5% discount band generated approximately: Revenue: **₦28,320,400,** Profit: **₦9,771,300, and** Profit margin: **34.5%***

*The 5–10% band generated: Revenue: **₦9,937,800,** Profit: **₦2,827,200, and** Profit margin: approximately **28.45%***

*The 10–15% band generated: Revenue: **₦5,588,325,** Profit: **₦1,516,625, and** Profit margin: approximately **27.14%***

*The 15%+ band generated: Revenue: **₦2,833,200,** Profit: **₦475,300, and** Profit margin: approximately **16.78%***

Therefore, the data does not support the claim that increasing discounts improves profitability. Profit margin decreases substantially as discount levels increase.

4. The Online channel has the lowest return rate.

Verdict

Incorrect

Corrected answer / result

In-store channel has the lowest return rate at approximately 6.44%, followed by Online channel at approximately 9.70%, while WhatsApp channel has the highest at approximately 11.90%.

Explanation / evidence

The In-Store channel records 202 transactions and 13 returns with 6.44% of return rate, Online channel records 134 transactions and 13 returns with 9.70% of return rate, and WhatsApp channel records 84 transactions and 10 returns with 11.90% of return rate.

Therefore, the original dashboard's claim is incorrect. Online actually has the highest return rate among the three channels.

5. August is the strongest sales month.

Verdict

Incorrect

Corrected answer / result

May is the strongest sales month based on revenue.

Explanation / evidence

May generated: ₦7,008,500 revenue, ₦2,053,400 profit, 55 transactions, and 162 units. While August, in comparison, generated: ₦3,167,425 revenue, ₦1,008,225 profit, 33 transactions, and 94 units

May therefore significantly outperformed August on both revenue and profit.

6. Lagos is underperforming compared with other markets.

Verdict

Incorrect

Corrected answer / result

Lagos is the strongest performing market in the analyzed dataset.

Explanation / evidence

Lagos recorded: ₦13,692,700 revenue, ₦4,579,900 profit, 373 units, 124 transactions, and 33.45% profit margin.

It ranked above Kano, Ibadan, Port Harcourt, and Abuja on the key performance measures analyzed. Therefore, the claim that Lagos is underperforming is contradicted by the corrected analysis.

7. The reported Revenue field can be used directly for decision-making.

Verdict

Incorrect

Corrected answer / result

The reported Revenue field should be validated before being used for decision-making.

Explanation / evidence

The independently calculated revenue formula was: $Revenue = Units \times Unit_Price \times (1 - Discount)$.
*The formula matched **398 out of 420 transactions**, approximately **94.8%** of the non-missing records. However, **22 transactions** had discrepancies. Because revenue is used to evaluate sales performance, products, markets, and profitability, these discrepancies should be investigated and corrected before relying on the field for management decisions.*

For the corrected analysis, Revenue_Final_NGN was used while the original reported revenue was retained for comparison.

8. The dashboard is sufficiently reliable for management to act on without further data cleaning.

Verdict

Incorrect

Corrected answer / result

The original dashboard should not be relied upon for management decisions without further data cleaning and validation.

Explanation / evidence

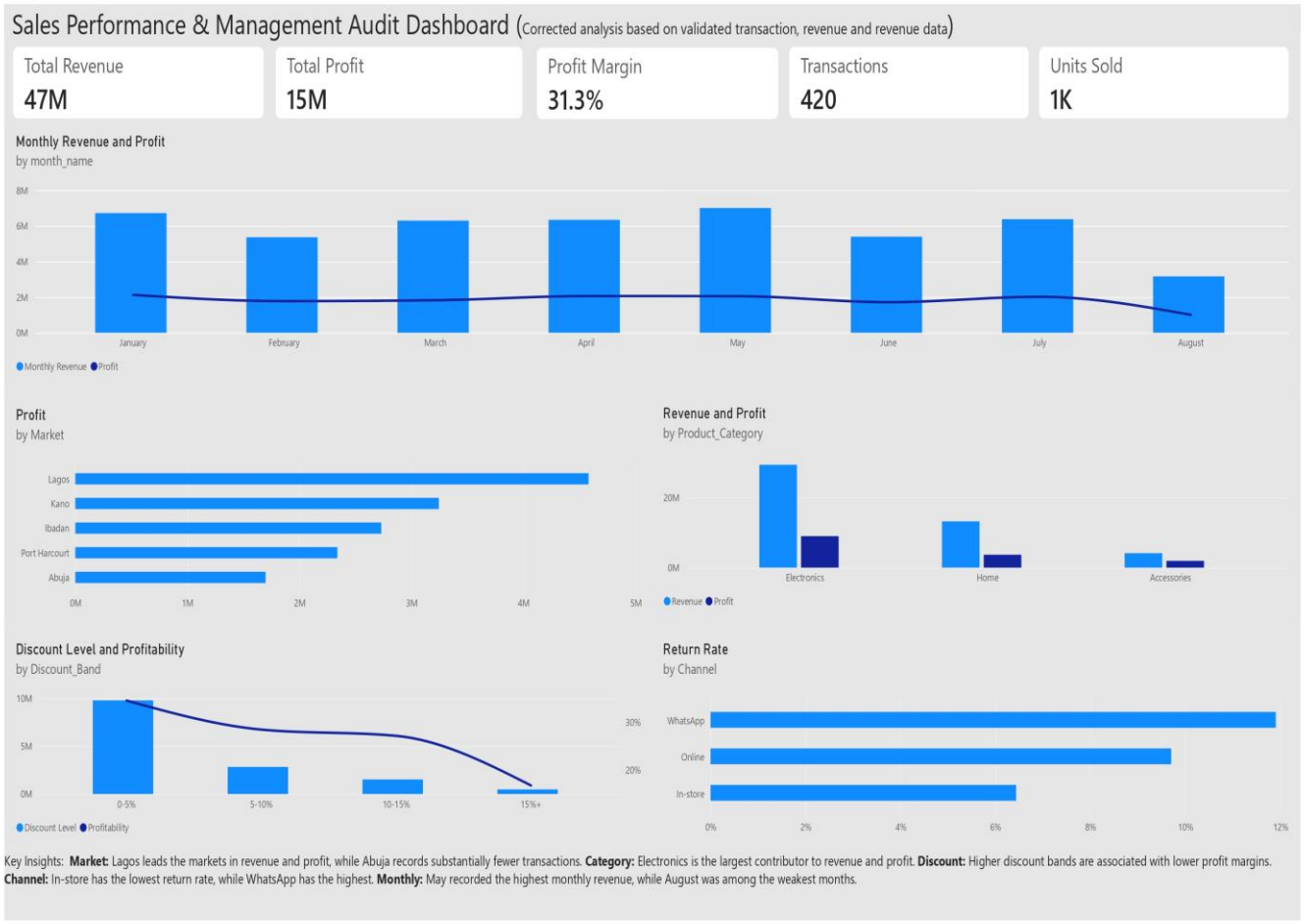
The analysis identified several data-quality problems, which including: 12 duplicate transaction records, inconsistent categorical values, missing cost per unit values, and 22 revenue calculation discrepancies. These issues can affect transaction counts, revenue, profitability, return rates, and comparisons between markets, categories, channels, and months.

Therefore, the data required cleaning and validation before the dashboard could be considered reliable for decision-making.

Part C - Corrected Dashboard

Insert a clear snapshot of your corrected dashboard below. The dashboard may be created in Power BI or Excel.

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Key Insights: **Market:** Lagos leads the markets in revenue and profit, while Abuja records substantially fewer transactions. **Category:** Electronics is the largest contributor to revenue and profit. **Discount:** Higher discount bands are associated with lower profit margins. **Channel:** In-store has the lowest return rate, while WhatsApp has the highest. **Monthly:** May recorded the highest monthly revenue, while August was among the weakest months.

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Data Quality and Audit Dashboard

Records Before Cleaning	Records After Cleaning	Duplicate Records Removed	Revenue Discrepancies	Revenue Formula Match
432	420	12	22	94.8%

Issue	Finding	Action
Duplicate Transaction	12 duplicate records were identified	Removed after confirming they contained identical data
Inconsistent Categories	State/Category Variations	State and product-category values were standardized to prevent duplicate categories.
Missing Costs	Missing Cost Per Unit Values	Missing Cost Per Unit values were mapped using the corresponding product's established cost.
Revenue Discrepancies	22 Records	Revenue Recalculated
Revenue Validation	398 of 420 records matched the formula, while 22 records required correction.	Revenue was independently recalculated

Part D - Final Analysis Questions

1. Which conclusion in the original dashboard was the most misleading, and why?

Answer

The conclusion that higher discounting is improving profitability was particularly misleading.

Explanation / evidence

*The corrected analysis shows the opposite pattern. The 0–5% discount band had a profit margin of approximately **34.5%**, while the 5–10% band had approximately **28.45%**, also 10–15% band had approximately **27.14%** and the 15%+ band had approximately **16.78%**. Therefore, higher discount levels were associated with substantially lower profit margins in this dataset.*

This conclusion could be particularly misleading because it could encourage management to increase discounts under the assumption that doing so improves profitability.

2. What changed most materially after you cleaned and corrected the data?

Answer

The reliability of revenue and profitability analysis changed most materially after cleaning and correcting the data.

Explanation / evidence

The original dataset contained duplicate transactions, inconsistent categorical values, missing cost values, and 22 transactions where reported revenue did not match the validated revenue formula. After removing duplicate records, standardizing categories, resolving missing product-based costs, and using the validated revenue calculation, the resulting data provided a more consistent basis for comparing profitability across markets, products, channels, and months.

3. If management had acted on the original dashboard, what decision could have created the greatest business risk?

Answer

Increasing discounts based on the assumption that higher discounting improves profitability could have created significant business risk.

Explanation / evidence

*The corrected analysis shows that higher discount bands were associated with lower profit margins. Profit margin decreased from approximately **34.5% at 0–5% discounting** to approximately **16.56% at 15%+ discounting**.*

If management responded to the original claim by increasing discounts without considering their effect on margins, the business could reduce profitability even if sales volume or revenue increased.

4. Based on your corrected analysis, what is the most important driver of business performance in this dataset?

Answer

Sales volume and product mix appear to be important drivers of overall business performance, with Electronics making the largest contribution to revenue and profit.

Explanation / evidence

*Electronics generated approximately **₦29.35 million in revenue and ₦8.99 million in profit**, substantially more than the other product categories. It also recorded the highest number of units and transactions among the categories analyzed.*

*The analysis also shows that discount levels are strongly associated with profitability, with higher discounts corresponding to lower profit margins. Therefore, management should consider both **product mix/sales volume and pricing/discount strategy** when evaluating performance.*

5. What three actions should management take next based on the evidence from your corrected dashboard?

Answer

1. *Review and optimize the discount strategy.*
2. *Investigate the performance gap between markets, particularly Abuja, while maintaining the strong performance of Lagos.*
3. *Monitor product-category performance and channel return rates more closely.*

Explanation / evidence

1. ***Review and optimize the discount strategy.***
   The corrected analysis shows that higher discount levels are associated with lower profitability. The 0–5% discount band generated approximately ***₦9.77 million in profit with a 34.5% profit margin****, while the 15%+ discount band generated approximately* ***₦475,300 in profit with a margin of about 16.56%****. Management should therefore review high-discount transactions and determine whether the additional sales generated justify the reduction in margin.*
2. ***Investigate differences between markets.***
   Lagos is the strongest-performing market, generating approximately ***₦13.69 million in revenue and ₦4.58 million in profit****, while Abuja generated approximately* ***₦6.68 million in revenue and ₦1.70 million in profit****. Management should investigate why performance differs between markets by examining product mix, customer behavior, pricing, sales channels, and demand before deciding on specific interventions.*
3. ***Monitor product-category performance and channel returns.***
   Electronics generated approximately ***₦29.35 million in revenue and ₦8.99 million in profit****, making it the largest contributor among the categories analyzed. At the same time, return rates differed across channels:* ***In-store = 6.44%, Online = 9.70%, and WhatsApp = 11.90%****. Management should continue monitoring category profitability while investigating why return rates are higher in some channels, particularly WhatsApp.*

Submission reminder: Complete all sections, insert a readable corrected-dashboard snapshot, ensure your written figures match the dashboard, and keep your Power BI or Excel working file available if verification is requested.